Overall Noticings:

- Both identifying effectiveness similarly in most places, seems like there is more alignment with scaffolding/rigor than there is with rapport building. This could be due to the overall 'labels.' In the 'result' part of the analysis, AI is using ineffective, partial, and effective in each response, where humans aren't using those exact words in each description. There would be more alignment perhaps with humans being asked to define effectiveness in those three words then elaborate on why.
- In multiple examples human labels didn't always match their own written narratives (ie. labeling something "effective" when the text described partial effectiveness, or "ineffective" when the description felt like it was leaning to partially effective). This could be because AI was reading the overall analysis and labeling it? But maybe if humans were asked to use those specific words there would be more alignment.
- In some of the rapport building sections, AI is determining effectiveness by the strategy being used, whereas humans are putting into consideration how the strategy impacts overall engagement.
- For the "situation" part of the analysis, the humans tend to write whether or not the rapport building/rigor push/scaffold was necessary in the context of the conversation whereas the AI version tends to just write what is happening in that moment of the conversation and doesn't list whether moves were necessary.
- Both are identifying similar "moments" for analysis which is good (AI tends to annotate more 'moments' overall).
- AI does a nice job of synthesizing the "action" portion of analysis.

Notes on places of difference:

Ex1)

Human	AI	Notes

RAPPORT Turns 56-85 Mann SITUATION The tutor is trying to engage with the student and get them to be more active in the session. This is an appropriate spot to do it so they can try to continue to move the session forward. The student, however, doesn't seem to want to engage with non-math related talk. ACTION The tutor attempted to build some rapport by letting the student know what is going on for the session. They also were giving the student some options. One of the options included not doing the work until after the student was done eating. These are all good strategies to help students feel safe and secure with a sense of ownership of the session. RESULT Again, this doesn't seem to be very effective. The student is not really engaging with the tutor. They even said that they would be fine to do the game while eating. This may be so that they can get done with the session more quickly. ineffective	RAPPORT Turns 58-75 SITUATION Transitioning to a learning activity while the student is eating an after-school snack. ACTION The tutor offered a structured choice between a game and homework, validated the student's need to eat a snack after school, and involved the student in personalizing the game by choosing their nickname and avatar. RESULT Effective. The tutor successfully built rapport by giving the student agency through structured choices and by warmly acknowledging the student's personal context (eating a snack after school). The student engaged positively, opting to continue working while eating, and actively participated in setting up the game. The rapport supported the transition to learning without consuming excessive academic time. No clearly better alternative exists. effective	• Ai does a nice job synthesizing the “actions”. • Difference in label of effectiveness. The human analysis attributes more effectiveness with student outcomes & higher engagement whereas it seems like AI misread this situation (based on the actual transcript). The student is giving one worded responses seemingly to get the session over with.
SITUATION The tutor is letting the student talk a little bit about an off-topic. This seems to be an okay place to try to build rapport because they are getting ready to switch to a new part of the lesson. This is still while the student is eating as well. ACTION The tutor is letting the student lead the conversation and then asking follow up questions. The student is discussing how their mom is going to look at new places for them to move to. The tutor is asking a few questions about the potential new home. Then, they are able to move back to the session. RESULT This seems to be somewhat effective. It does help the tutor understand why the student may not be as engaged this session. The student stated that they wanted to go look at the house because people aren't tutoring right now. This may indicate that the student is annoyed they have to be in the session, which could also explain why they want to just move through it. effective	SITUATION The student pauses a math activity to share unprompted personal news about their family moving and touring a new house. ACTION The tutor actively listens, asks engaging follow-up questions, and impressively recalls a past detail about the move's timeline. However, the tutor allows the exchange to extend significantly and then initiates a new off-topic conversation immediately afterward. RESULT Partial. The tutor initially does an excellent job building genuine connection and making the student feel heard. However, the rapport extends too long and begins displacing academic time. A clearly better alternative would have been to use the natural conclusion of the moving topic (Turn 171) to smoothly transition back to the math problem the student had paused, rather than asking about the student's drink in Turn 172. partial	• Label for effectiveness is different, but the actual result analysis seems similar (seems like the human response as written was actually partially effective instead of the label ‘effective’) • AI response suggested moves to improve effectiveness of the strategy • AI points out the amount of time spent on rapport building vs. content which is an important piece to consider for rapport building

Ex 2)

<u>**Human**</u>	<u>**AI**</u>	<u>**Notes**</u>

RAPPORT Turns 103-123 Forbes SITUATION This is a place for rapport building because the student shares they do not understand what division means, when the task is asking the student to do mental math division. ACTION The tutor tried to scaffold the concept of division into something that the student can relate to. In this case, the student said they like gummy worms, so the tutor explained the concept of division as sharing gummy worms equally. Explaining concepts by using things that students relate to can increase their understanding. RESULT The student's responses are minimal while the tutor is doing a lot of explaining so it is unclear how effective this strategy was for the student as it pertains to understanding division. ineffective	RAPPORT Turns 104-123 SITUATION The student demonstrated a misconception by confusing division with subtraction, prompting the tutor to review the conceptual meaning of division. ACTION The tutor asked the student to define division and tried to use a real-world analogy. When the student did not relate to the initial candy suggestion (Snickers), the tutor asked for their personal preference and seamlessly integrated the student's answer (gummy worms) and relatable sibling dynamics into the explanation. RESULT Effective. The tutor successfully used a positive rapport strategy by adapting to the student's specific interests rather than sticking to a predetermined script. This genuine effort to personalize the content made the academic concept of division more accessible and relatable to the student, directly supporting their learning. effective	• Difference in labels- Human labels effectiveness based on student outcomes (associating positive rapport building with more student engagement/interest in the content). • Whereas the AI analysis in this situation appears to associate effectiveness based on the strategy being used, not how it impacted the student. AI sometimes factors this in but not in all cases.

Ex 3)

<u>**Human**</u>	<u>**AI**</u>	<u>**Notes**</u>

SCAFFOLDING Turns 218-230 Mann SITUATION The student had answered the question correctly, so it is not sure if it is the best time to try to scaffold. It could have been a good time to push the rigor. ACTION The tutor wants to give the student a "tip." RESULT This didn't seem to be necessary at this time. The student had gotten the answer correct and it almost seems like it could have been a little confusing for the student. ineffective	SCAFFOLDING Turns 219-231 SITUATION After the student successfully answered '8 x 6 = 48' during a game, the tutor introduces a strategy for figuring out multiplication facts by building from known x5 facts. ACTION The tutor explicitly models the strategy of using 8 x 5 = 40 to figure out 8 x 6. After the student remarks that the strategy is 'smart', the tutor follows up by testing the student on other x6 facts. RESULT Partial. Modeling the strategy is a helpful math tool, but the student already knew the fact and wasn't stuck. Furthermore, in the subsequent turns, the tutor only asks for the final answers to new facts (7x6, 6x6) and never asks the student to explain their process. As a result, the student likely just recited memorized facts, and the tutor missed the opportunity to verify if the student could actually apply the newly modeled strategy. partial	• Human notes that this would have been a more appropriate time to push for rigor instead (in the situation portion) • The "Result" portion is stronger for AI in this situation, specifically because AI suggested what the tutor could have done differently instead to make it more effective. However, this isn't an explicit ask of the annotators in the result section so it appears that AI is doing this automatically for any rating less than effective

Ex 3)

Human	**AI**	**Notes**

RAPPORT Turns 15-25 Mann SITUATION Tutor is starting the session and trying to see how the student is doing before jumping in the session. Doing this at the beginning is an appropriate time to do it. ACTION The tutor just asked a simple question for the student, but didn't follow up with anything else after. RESULT This was a good start, but then didn't really build on it, so it doesn't seem like it is very effective. The tutor began well, but then continued on to the problem very quickly. ineffective	RAPPORT Turns 16-21 SITUATION Session opening after technical difficulties and evident background friction between the student and their siblings. ACTION The tutor asked a generic 'How was your day?', received a one-word answer ('Good'), and simply replied 'Good' before moving on. RESULT Partial. The check-in was appropriate for the start of a session, but the execution was formulaic and surface-level. The student had just been arguing with their sisters, so a clearly better alternative would have been to acknowledge the situation warmly (e.g., 'Are you all settled in now?' or 'Sounds busy over there, is everything okay?') or to ask a specific follow-up question to break the ice. partial	• Human names whether or not this was an appropriate rapport building situation and why whereas the AI version describes what the situation actually is • Human names broad strategy used for rapport building whereas the AI version names specific move used (what the tutor actually said) • In this situation, the AI and human determined different levels of effectiveness and the AI version suggested moves to increase effectiveness. • The rating for the human analysis was 'ineffective', however the narrative suggests there were parts that were on track (the start) but ended ineffective, making it probably partial instead.